PERSONAL INFORMATION MANAGEMENT (PIM) SYSTEM

Line Coverage Report

Document	Line Coverage Report
Version	1.0
Date	8 November 2026
Group	Group [number]
Status	Final

Group members

Name	Student ID
[Member 1 name]	[Student ID]
[Member 2 name]	[Student ID]
[Member 3 name]	[Student ID]
[Member 4 name]	[Student ID]

1 Summary

The unit tests in `tests/unit` execute **800 of the 800 countable lines** in the `model` package, a line coverage of **100.0 %**. Every file of the package is fully covered.

File	Countable lines	Lines executed	Line coverage
`model/criterion.py`	247	247	100.0 %
`model/pim.py`	91	91	100.0 %
`model/pimfile.py`	83	83	100.0 %
`model/pir.py`	379	379	100.0 %
Total	**800**	**800**	**100.0 %**

2 Method

The report is produced by `coverage_report.py` at the root of the source code. It uses only the Python standard library:

- The standard `trace` module runs every test in `tests/unit` and records which lines of `model/` were executed.
- A line is *countable* when a Python statement starts on it. Docstrings, comments, blank lines, and the continuation lines of a statement that spans several lines are not counted.
- `model/__init__.py` only re-exports names from the other modules, so it is excluded.
- The third-party `coverage.py` tool is not used. Its totals can differ slightly because it counts lines in its own way.

To reproduce the figures, run this command at the root of the source code:

```
python3 coverage_report.py
```

The command rewrites `coverage.txt` and exits with status 1 if a unit test fails or any countable line of `model/` is not executed.

3 Report output

The full text of `coverage.txt`:

```
Line coverage for model/ (stdlib trace, tests/unit)

Method: the standard-library ‘trace‘ module runs every test in tests/unit
and records which lines of model/ executed. A line is countable when an AST
statement starts on it; docstrings, comments, blank lines, and continuation
lines of a multi-line statement are not counted. model/__init__.py holds
re-exports only and is excluded. Third-party coverage.py is not used.

criterion.py: 247/247 lines (100.0%)
pim.py: 91/91 lines (100.0%)
pimfile.py: 83/83 lines (100.0%)
pir.py: 379/379 lines (100.0%)

TOTAL model/: 800/800 lines (100.0%)
```